5 Installation and dismantling *RR 450*

5.1 Checking the delivery content and condition

The RR 450 is mostly pre-assembled at delivery. The side columns are fixed together with the top roll cover, the motor cover (if included in delivery), the fixing material and the control unit, to a transport pallet (made of wood). The door must be checked for possible transport damages and completeness – compare with order/delivery papers.

5.2 Preparations for installation work

Check the place of installation with regard to fixing points and set-up conditions. Possibly a supporting construction or other required fixing possibilities must be prepared. The head plates should be screwed onto a steel support or a concrete element.

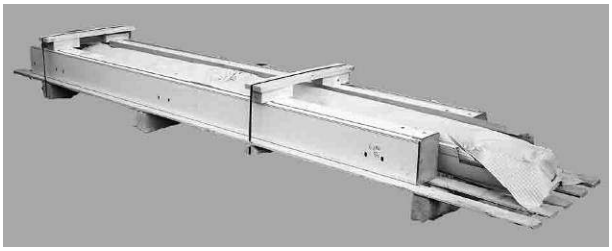


Isolated panel walls, rider bricks and lightweight construction walls must be checked for suitability of installation.

Measure the light dimensions (width and height) of door opening and check with order papers.

The site of installation must be secured against pedestrian and vehicle traffic.

Before start of installation the plastic sheets of the packing must be removed. Afterwards the covers of the side columns and the control unit must be removed from the pallet and laid down in sufficient distance of the door opening.



1. The door shipment should be checked for transport damages and completeness – compare delivery / order papers.



2. The floor must be checked for even level. This may be done with a laser pointer or a hose level.



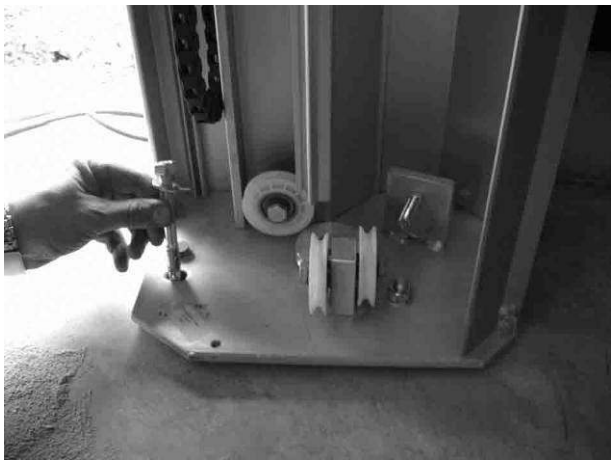
3. At slanting surface of the floor the side column must be firstly installed on the higher side. (difference in height max. 3mm)



4. Align first side column exactly vertically by means of a water level.



5. Fix the head plates with 2 screws onto the wall.



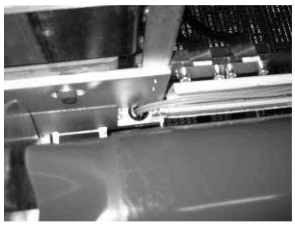
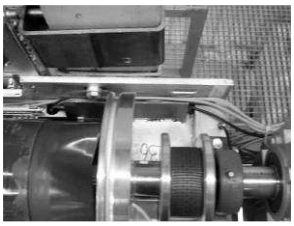
6. Fix the bottom plate with 2 screws.



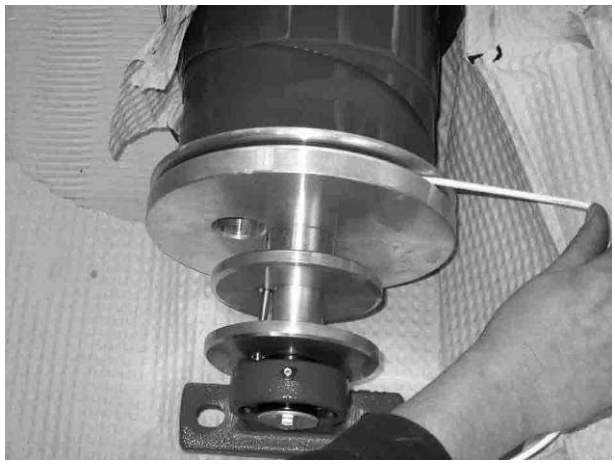
7. Set up the second side column.
Difference in height between the side columns **max. 3 mm**. If necessary the lower side must be lined.



8. Align the second side column to the exact light width and fix as described in point 5. and 6. **(daylight width ± 2mm)**

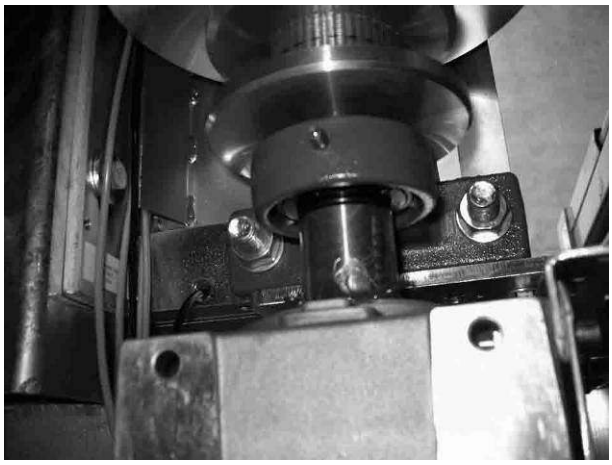

9. Install the aluminum U profile for the cable wiring at both head plates and lay the cable.



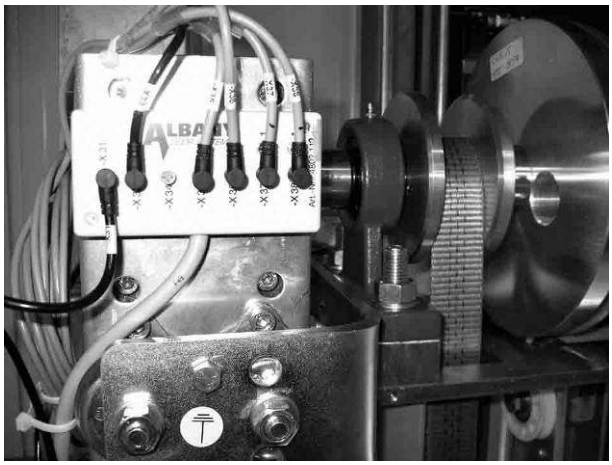
10. Pull the rope through the opening of the winding disc, knot together and make **two** pre-winding layers opposite the curtain direction.



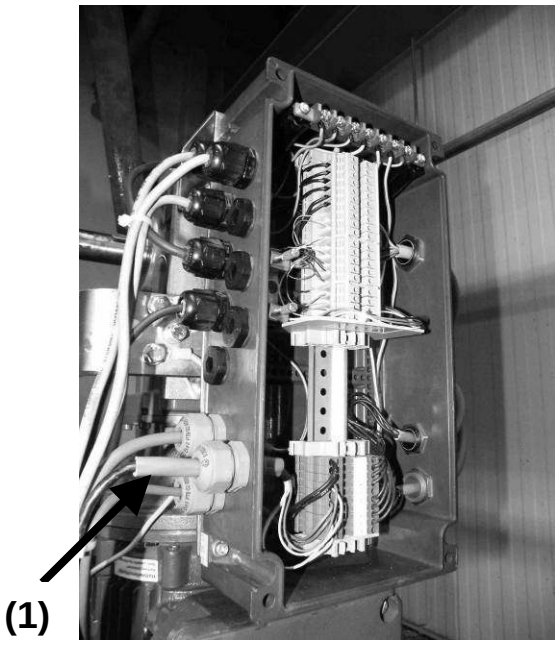
11. Lift the top roll with a crane or forklift. With small doors the top roll may be lifted by hand.



12. Fix the bearings onto the head plates at the prepared points.

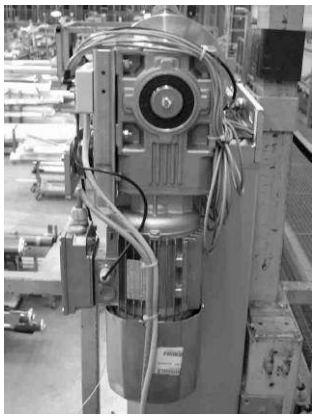


13. Put the motor onto the drive shaft. Pay attention to the feather key connection. Connect the cables to the splitter box according to the numbers.

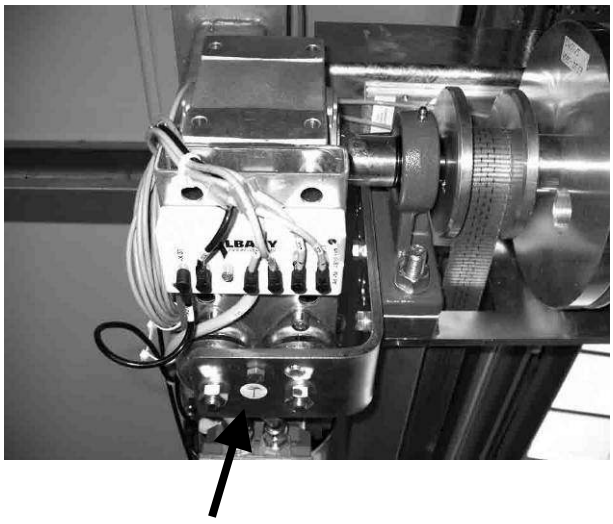


14. RR450Ex: Mark cable from safety edge at terminal box with blue shrink tube.
Stationary photocell, safety edge and reference limit switch through blue threaded joints

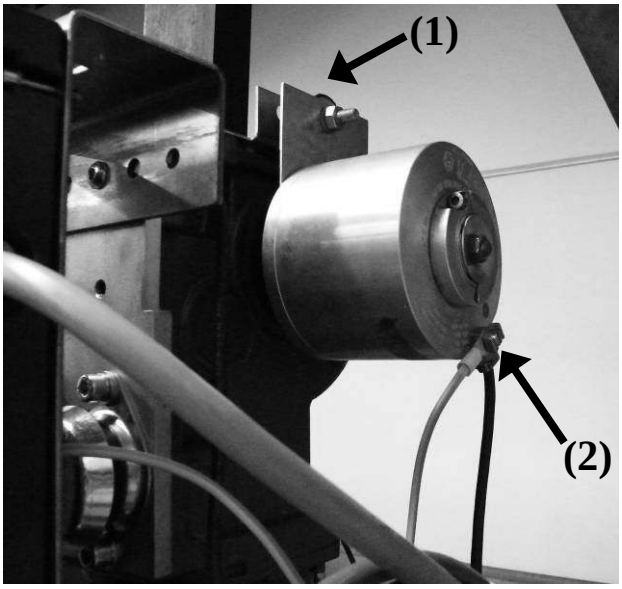




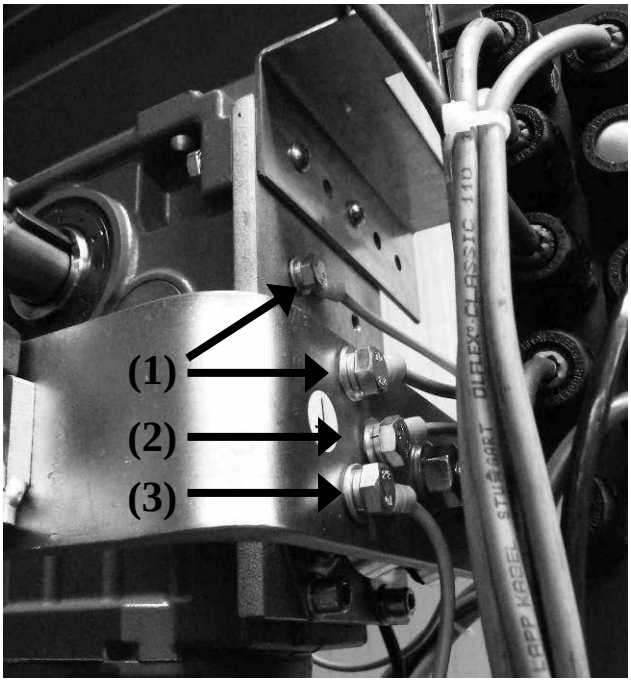
15. Pull the metal wire through the bore holes of the head plate. The end is led through the brake release lever and fixed at the cover plate. Extra cable should be wound up and connected to the motor.



16. Fix the torque support with 2 screws at the head plate and screw together with the motor. **Ground door to building at torque arm (min. 6 mm²)**



17. RR450Ex: Installation encoder
(1) with rubber block
(2) connect ground from torque arm



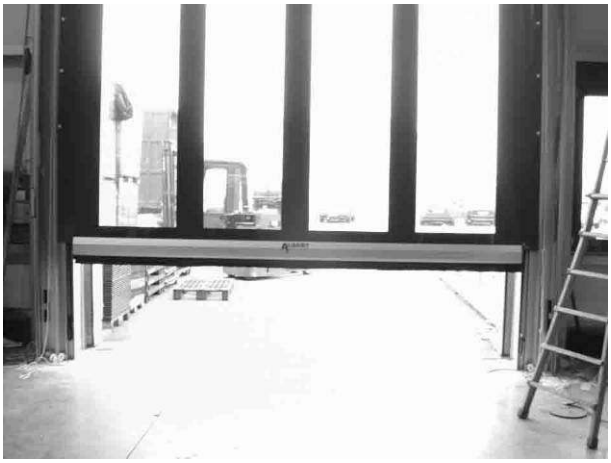
18. RR450Ex: Ground drive unit (1) and encoder (2)at torque arm; Ground door to building (min. 6 mm²) (3)



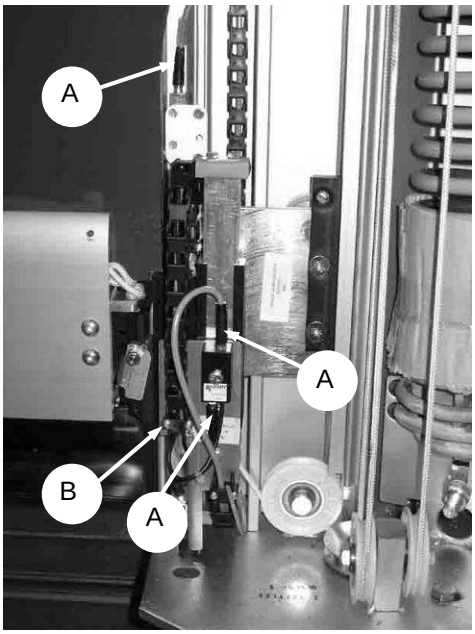
19. Put on the ropes for the curtain tension according the drawing at the side columns.



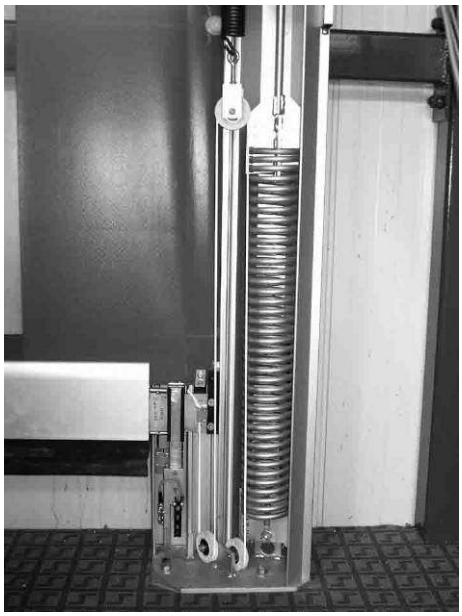
20. Using the emergency lever. Let the door down until the bottom of the curtain reaches the same height of the emergency lever.



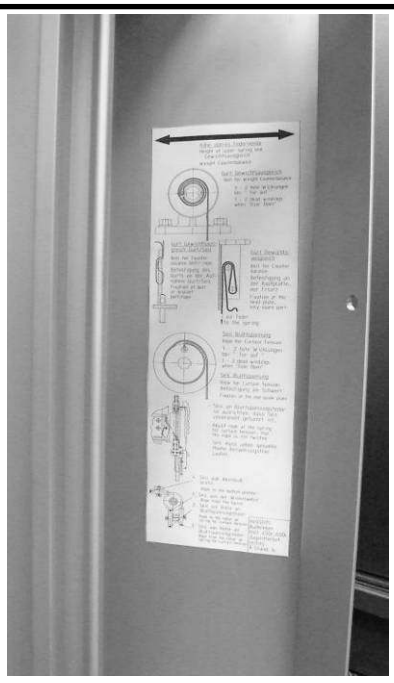
21. Slip the bottom profile onto the bulge of the door curtain and center. Connect guide blocks without sliding guidance to the bottom profile.



22. Install the flexible cable duct between side column and end guiding plates. Connect wiring **A,** and earth connection **B**, according to the drawing.

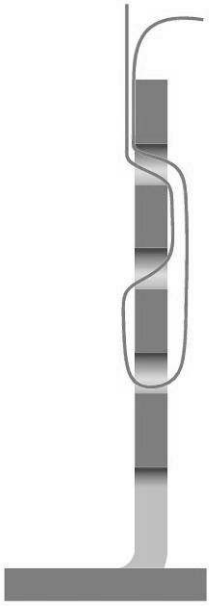


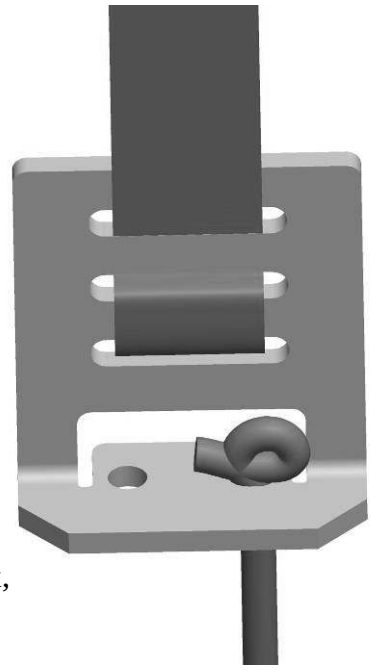
23. Thread ropes through pulleys according to drawing located in the side columns.



24. The door must be in the open position to install the counterbalance system. Connect the belts to the drums according to the drawing in the side columns.

Drawn cut of
bracket belt-rope

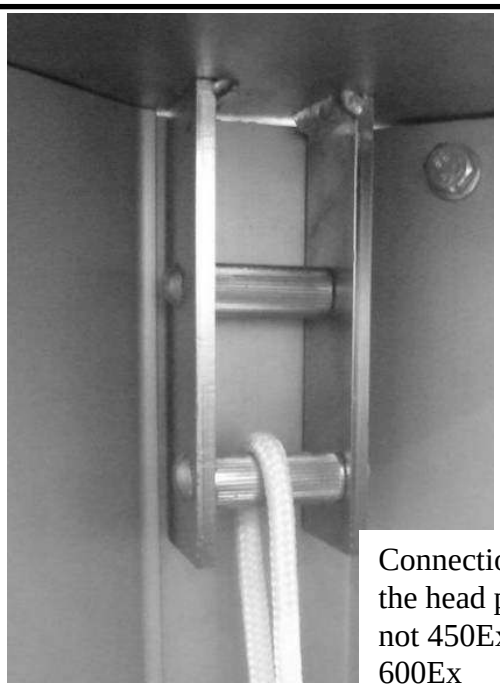




not 450Ex,
600Ex

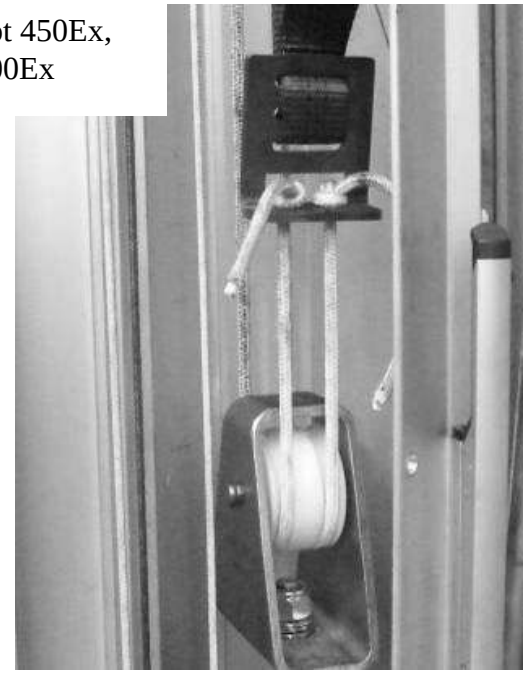
25. Use a hoist to pre tension the spring to the mark in the side columns.
Fasten the bracket belt-rope at the blue belt as shown in sketch. (not 450Ex, 600Ex)

26. Make a knot at the end of the yellow rope **(**Ø 7mm) (picture). Pull the rope through the hole in the bracket belt-rope and then under the pulley of the spring seat top. (not 450Ex, 600Ex)



Connection at
the head plate,
not 450Ex,
600Ex

not 450Ex,
600Ex



27. Direct the yellow rope (Ø 7mm) over the bolt in the head plate and then back under the pulley of the spring seat top and through the 2nd hole in the bracket belt-rope. (not 450Ex, 600Ex)

28. The following must be done at open door position and pre tensioned springs. Position the bracket belt-rope 150mm above the spring seat top. Make a 2nd knot. Advice: The rope will elongate under tensile load. (not 450Ex, 600Ex)



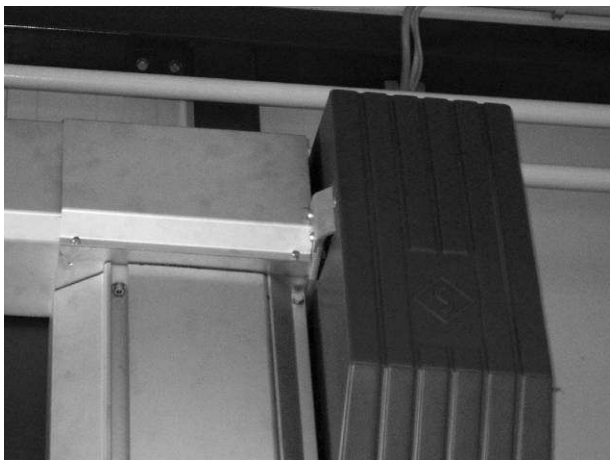
not 450Ex,
600Ex



29. Take care that you set up the value that is given in picture 26. If the distance is much more than 150mm, the bracket belt-rope maybe hit the head plate, this can cause damage. (not 450Ex, 600Ex)

30. Screw together the top roll cover and covers with the head plates.

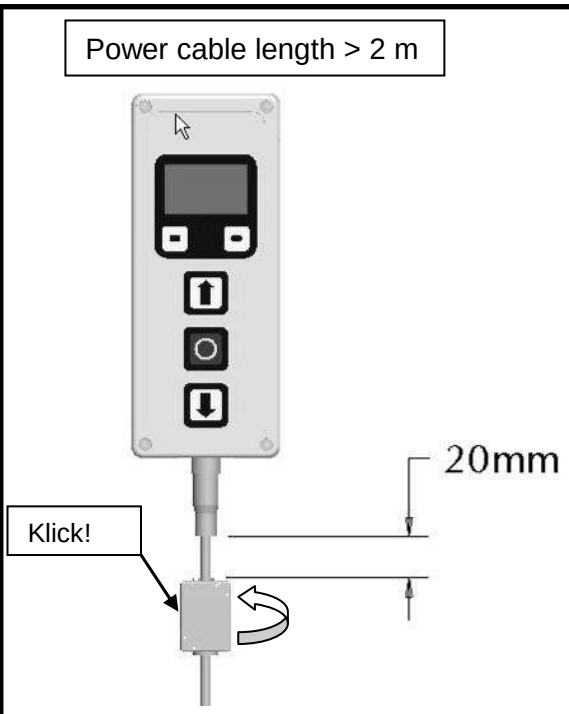




31. Screw on the endcaps as noted.

32. Screw the motor cover onto the designated fixiation.





33. Screw cover profile together with bottom and head plates.

After testing the safety devices and the initial operation/test run of the door the installation work is finished.

34. After testing the safety devices and the initial operation/test run of the door the installation work is finished.

The dismantling of the door is made in opposite direction to the installation.

The packing material of the door may be returned to the manufacturer. Defective door components should be disposed at site according environmental regulations.

Defective electronic components should be disposed as special waste.